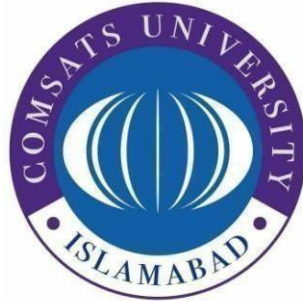


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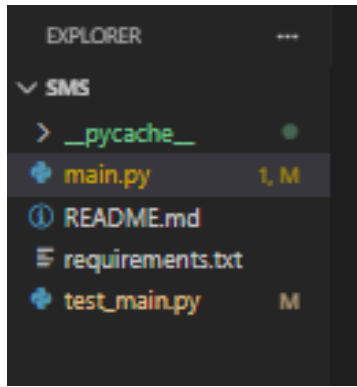
Course	SQE Lab
Instructor	Ma'am Munazza
Title	Semester-Project
Date	18-May-2026

SUBMITTED BY:

Reg. Numbers	FA23-BSE-011 FA23-BSE-025 FA23-BSE-028 FA23-BSE-043 FA23-BSE-045 FA23-BSE-053
Group Member	Sameer Ahmed Talha Ahmed Alishba Waseem Aima Jahangir Esha Zia Usama Saleem

Project : “ Student Management System”

Folder :



Main.py :

```
main.py
1  # main.py
2
3  class StudentManagementSystem:
4      def __init__(self):
5          self.students = []
6
7      # Normalize helper (avoids duplicate logic)
8      def normalize(self, name):
9          return name.strip().lower()
10
11     # Add Student
12     def add_student(self, name):
13         name = name.strip()
14
15         if name == "":
16             return "ERROR: Name cannot be empty!"
17
18         if self.normalize(name) in [self.normalize(s) for s in self.students]:
19             return f"WARNING: '{name}' already exists!"
20
21         self.students.append(name)
22         return f"SUCCESS: '{name}' added successfully!"
23
24     # View Students
25     def view_students(self):
26         if len(self.students) == 0:
27             return "No students found."
28
29         result = []
30         for i, s in enumerate(self.students, 1):
31             result.append(f"{i}. {s}")
32         return result
33
34     # Search Student
35     def search_student(self, name):
36         name = self.normalize(name)
37
38         for s in self.students:
39             if self.normalize(s) == name:
40                 return f"FOUND: {s}"
41
42         return "NOT FOUND"
```

```

44     # Delete Student
45     def delete_student(self, name):
46         name = self.normalize(name)
47
48         for s in self.students:
49             if self.normalize(s) == name:
50                 self.students.remove(s)
51                 return f"DELETED: {s}"
52
53         return "NOT FOUND"
54
55 # ===== MAIN MENU =====
56 if __name__ == "__main__":
57     system = StudentManagementSystem()
58
59     while True:
60         print("\n===== STUDENT MANAGEMENT SYSTEM =====")
61         print("1. Add Student")
62         print("2. View Students")
63         print("3. Search Student")
64         print("4. Delete Student")
65         print("5. Exit")
66
67         choice = input("Enter choice: ").strip()
68
69         if choice == "1":
70             name = input("Enter name: ")
71             print(system.add_student(name))
72
73         elif choice == "2":
74             result = system.view_students()
75             if isinstance(result, list):
76                 print("\n".join(result))
77             else:
78                 print(result)
79
80         elif choice == "3":
81             name = input("Enter name: ")
82             print(system.search_student(name))
83
84         elif choice == "4":
85             name = input("Enter name: ")
86             print(system.delete_student(name))
87
88         elif choice == "5":
89             print("Program exited.")
90             break
91
92         else:
93             print("Invalid choice!")

```

Test main.py :

```
test_main.py
1  # test_main.py
2
3  import unittest
4  from main import StudentManagementSystem
5
6
7  class TestStudentManagementSystem(unittest.TestCase):
8
9      def setUp(self):
10         self.system = StudentManagementSystem()
11         self.system.students = ["Ali", "Ahmed"]
12
13         # Test Add Student
14         def test_add_student(self):
15             result = self.system.add_student("Usama")
16             self.assertIn("SUCCESS", result)
17             self.assertIn("Usama", self.system.students)
18
19         # Test Add Duplicate
20         def test_add_duplicate(self):
21             result = self.system.add_student("Ali")
22             self.assertIn("already exists", result)
23
24         # Test Search Found
25         def test_search_found(self):
26             result = self.system.search_student("Ali")
27             self.assertIn("FOUND", result)
28
29         # Test Search Not Found
30         def test_search_not_found(self):
31             result = self.system.search_student("Zain")
32             self.assertEqual(result, "NOT FOUND")
33
34         # Test Delete Student
35         def test_delete_student(self):
36             result = self.system.delete_student("Ahmed")
37             self.assertIn("DELETED", result)
38             self.assertNotIn("Ahmed", self.system.students)
39
40         # Test Delete Not Found
41         def test_delete_not_found(self):
42             result = self.system.delete_student("Zain")
43             self.assertEqual(result, "NOT FOUND")
44
45
46     if __name__ == "__main__":
47         unittest.main()
```

Output :

```
PS C:\Users\alish\Desktop\SMS> python3 test_main.py
```

```
*****
-----
Ran 6 tests in 0.002s
```

```
OK
```

Snyk.io :

 **Today**

▼ **alishbawaseem-028/SQE-Project** main

Import triggered at 19:39:05

Project	Status	
Snyk Code supported files	✔ Project created	View project
requirements.txt	✔ Project created	View project

 **Code Analysis**

Overview

History

Settings

1. ● **Latest, 4 minutes ago, by snyk.io**

0 known issues

0

C

0

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2. ○ 21 minutes ago, by snyk.io

0 known issues

0

C

0

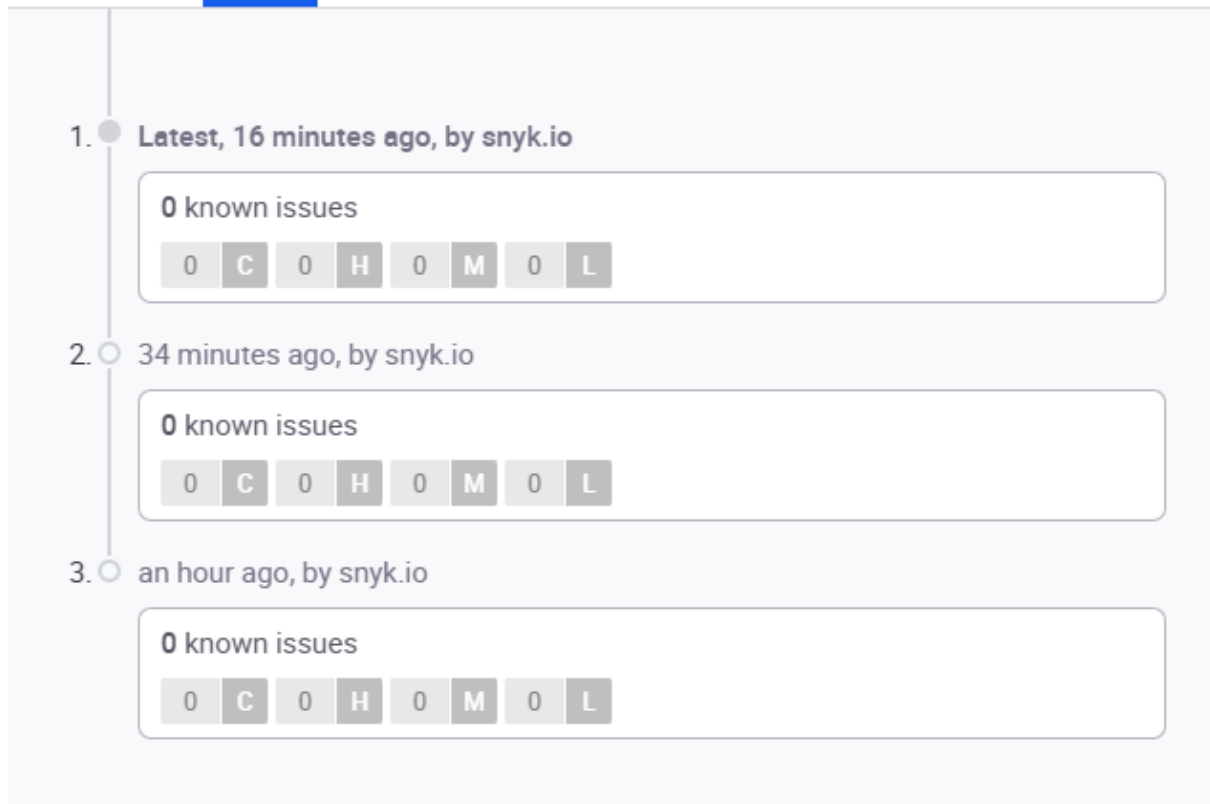
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L



Snyk Security Analysis Result

The Snyk scan of the project returned **0 security vulnerabilities** in project dependencies. This indicates that all installed packages are up-to-date and do not contain known security risks.

Security Issues (Design-Level / Logical Issues)

Even though no dependency vulnerabilities were found, the following **potential security risks** exist in system design:

1. No Authentication System

- Anyone can access and modify student records
- **!** Risk: unauthorized access

Fix: Add login system

2. No Role-Based Access Control

- All users have full permissions

Fix: Admin vs student roles

3. In-Memory Data Storage

- Data is lost when program closes
- **!** Risk: data integrity issue

Fix: Use database (SQLite / MySQL)

4. No Input Validation Layer

- System trusts raw user input

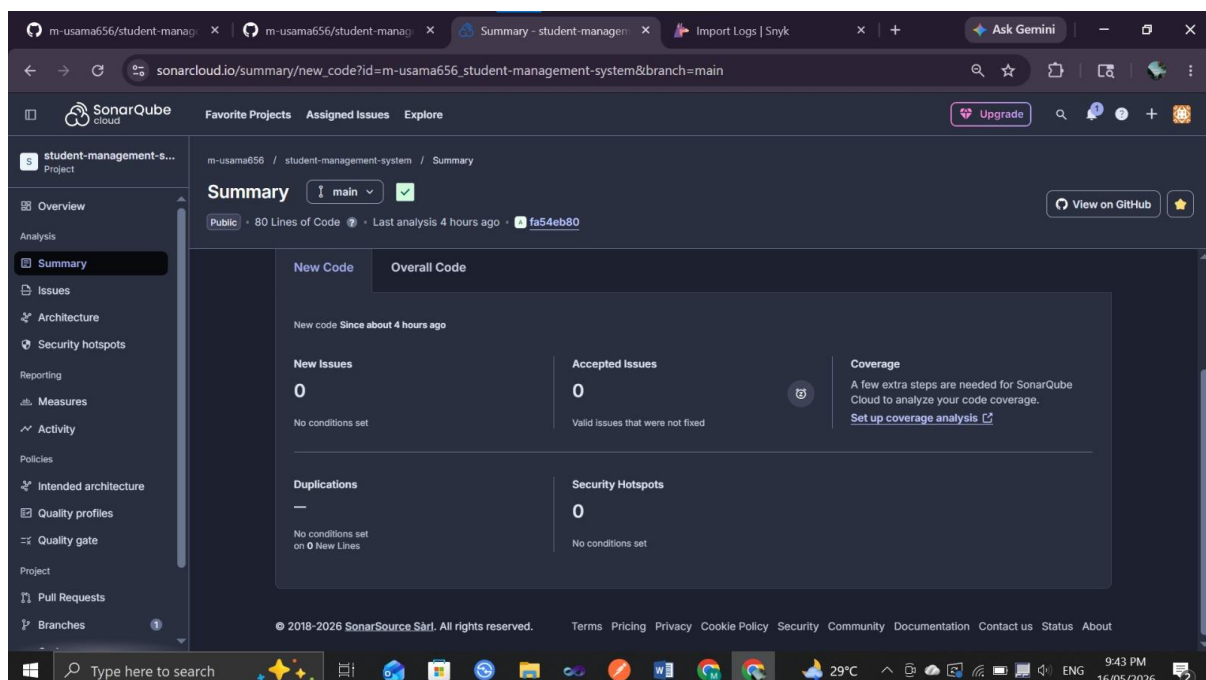
Fix: Add strict validation & sanitization

5. No Audit Logging

- No record of who added/deleted data

Fix: Add logging system

SonarQube :



sonarcloud.io/summary/overall?id=m-usama656_student-management-system&branch=main

SonarQube cloud

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student-management-s... Project

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Issues

Architecture

Security hotspots

Reporting

Measures

Activity

Policies

Intended architecture

Quality profiles

Quality gate

Project

Pull Requests

Branches

m-usama656 / student-management-system / Summary

Summary main

Public · 80 Lines of Code · Last analysis 4 hours ago · fa54eb80

View on GitHub

New Code Overall Code

Security 0 Open Issues A

Reliability 0 Open Issues A

Maintainability 0 Open Issues A

Accepted Issues 0

Coverage 0.0%

Duplications 0.0%

Security Hotspots 0

0 Open Issues

0 Open Issues

0 Open Issues

Valid issues that were not fixed

A few extra steps are needed for SonarQube Cloud to analyze your code coverage. [Set up coverage analysis](#)

No conditions set on 110 Lines

Type here to search

9:46 PM 16/05/2026

sonarcloud.io/project/issues?issueStatuses=OPEN%2CCONFIRMED&id=m-usama656_student-management-system

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Quality profiles

Quality gate

Project

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Branches

m-usama656 / student-management-system / Issues

Issues main

Filters

Software quality

Security 0

Reliability 0

Maintainability 0

Severity

Blocker 0

High 0

Medium 0

Low 0

Info 0

Code attribute

Last analysis had a warning [View warning](#)

Select issues for bulk actions

Change Select issues Navigate to issue 0 issues 0 effort

No Issues. Hooray!

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9:47 PM 16/05/2026

Bug Report :

Bug 1

Bug ID: SGMS_BUG_01

Bug Name: SonarCloud Analysis Warning / Missing Test Execution Logs

Description: The SonarCloud dashboard displays a banner stating "Last analysis had a warning." This indicates that while the static analysis completed, test execution reports or coverage metrics were missing/misconfigured during the build phase. Steps to Reproduce: 1. Open the SonarCloud project dashboard for student-management-system.

2. Observe the yellow warning banner at the top of the issues screen.

Expected Result: The analysis completes cleanly with zero execution warnings.

Actual Result: Dashboard displays "Last analysis had a warning" due to unindexed test executions.

Severity: Medium

Status: Fail

Bug 2

Bug ID: SGMS_BUG_02

Bug Name: Decoupled Test State in Unit Tests

Description: In test_main.py, the test cases perform actions directly on a local dummy list students = ["Ali", "Ahmed"] declared inside the test file itself instead of importing and testing the state changes of the actual students list inside main.py. This leads to deceptive "Passed" test outcomes that do not validate the core application state.

Steps to Reproduce: 1. Open test_main.py.

2. Run the test suite via pytest or python test_main.py.

3. Observe that tests pass despite never interacting with the application code inside main.py.

Expected Result: Test assertions should run against the functions and memory state imported directly from main.py.

Actual Result: Unit tests pass, but they are completely isolated from the real runtime file.

Severity: High

Status: Fail

GitHub :

